Kinetic Energy Driven Ferromagnetic Insulator

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20th 全国磁学理论会议

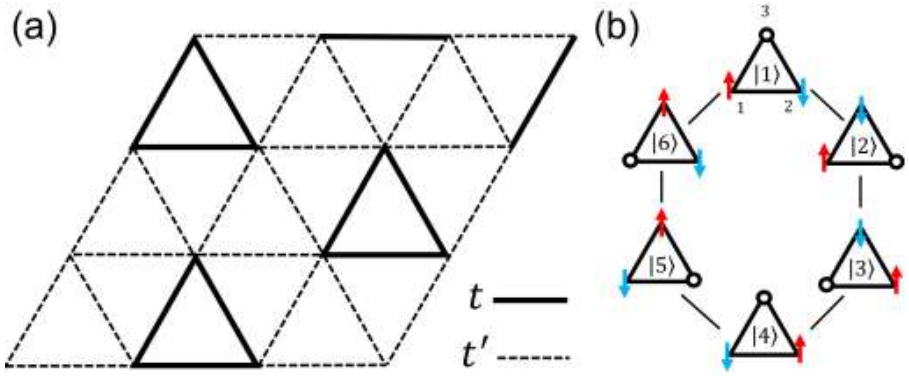
2026/5, Hangzhou

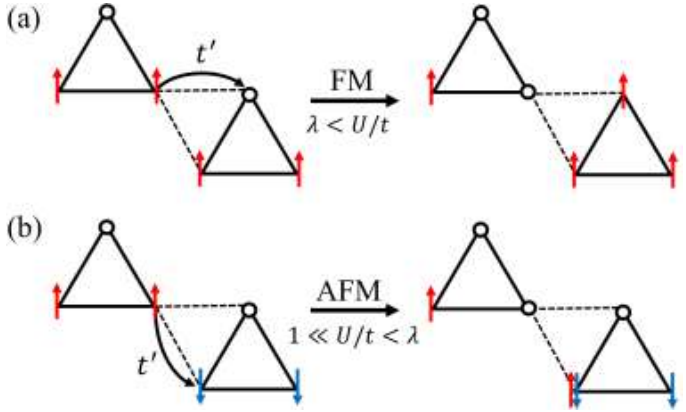
Phys. Rev. Lett. 136, 086504(2026)

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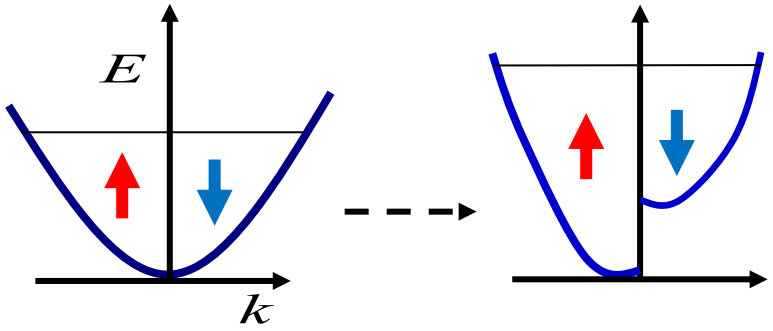


20th 全国磁学理论会议

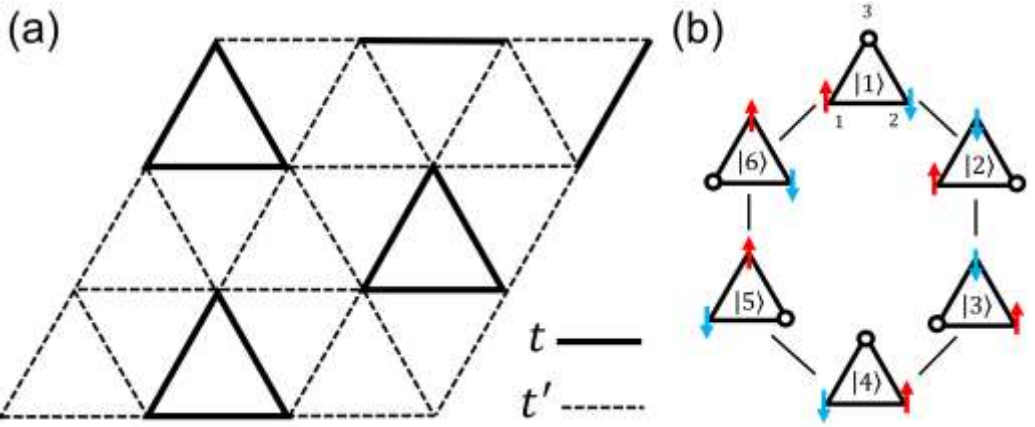
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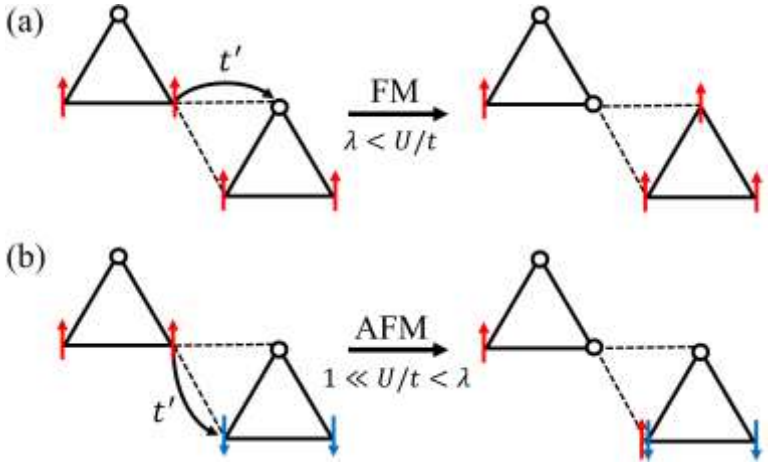
1.The origin of ferromagnetism – interplay between kinetic energy and interaction



2.From Nagaoka ferromagnetism to kinetic energy driven ferromagnetic insulator

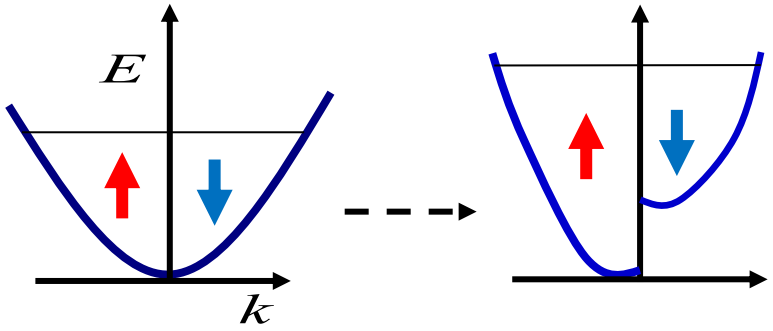


3.Competition between AFM and FM



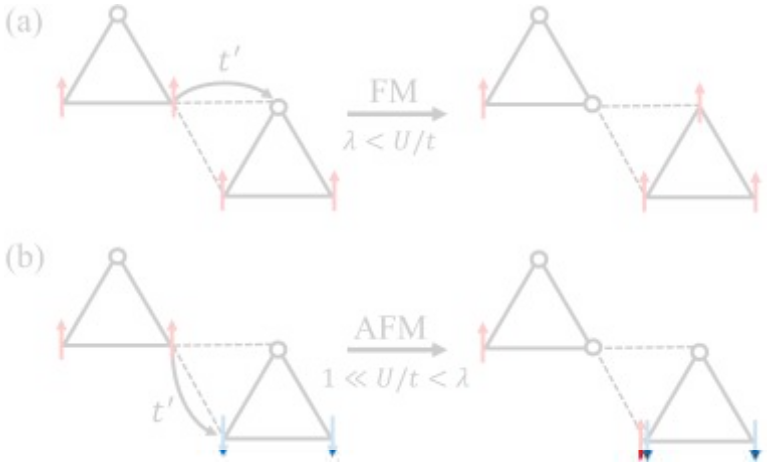
4. Extension

1.The origin of ferromagnetism – interplay between kinetic energy and interaction



2.From Nagaoka ferromagnetism to kinetic energy driven ferromagnetic insulator

3.Competition between AFM and FM



4. Extension

The origin of ferromagnetism

- Absence of magnetism in classic mechanics -- **Bohr-van Leeuwen Theorem**

$$H(r_i, p_i) = \sum_i \frac{\left(p_i - eA(r_i)\right)^2}{2m} + \frac{1}{2}\sum_{i,j} U(r_i, r_j)$$

$$Z[A] = \int \prod_{i=1}^{N} dr_i dp_i e^{-\beta H[A](r_i, p_i)} = Z[A=0]$$

$$\Rightarrow \quad M = -\frac{1}{\beta}\frac{d \ln Z}{d(\nabla \times A)} = 0$$

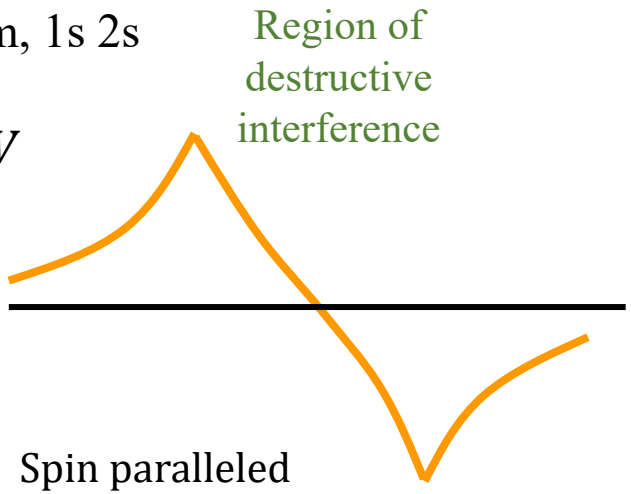
No magnetization even in a field!

- Spin, **Hund's rule** and ferromagnetism



Excited state of Helium, 1s 2s (triplet/singlet):

$$E_s - E_t \approx 0.8eV$$



Spin anti-paralleled

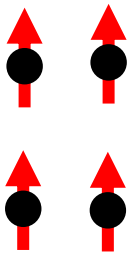
$$\psi_s \sim \left(\psi_1(r_1)\psi_2(r_2) + \psi_1(r_2)\psi_2(r_1)\right)\left(|\uparrow_1\downarrow_2\rangle - |\downarrow_1\uparrow_2\rangle\right)$$

Spin paralleled

$$\psi_t \sim \left(\psi_1(r_1)\psi_2(r_2) - \psi_1(r_2)\psi_2(r_1)\right)\left(|\uparrow_1\downarrow_2\rangle + |\downarrow_1\uparrow_2\rangle\right)$$

Locality v.s. itinerancy

- Ferromagnetism in insulator



$$H = \sum_{ij} J_{ij} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

$$J_{ij} = 2 \int \phi_i^*(r_1)\phi_j^*(r_2) \frac{e^2}{|r_1 - r_2|} \phi_i(r_2)\phi_j(r_1)$$

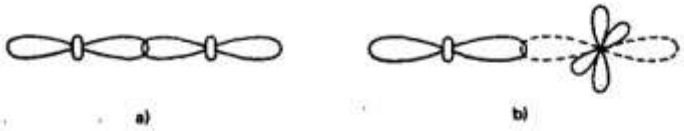
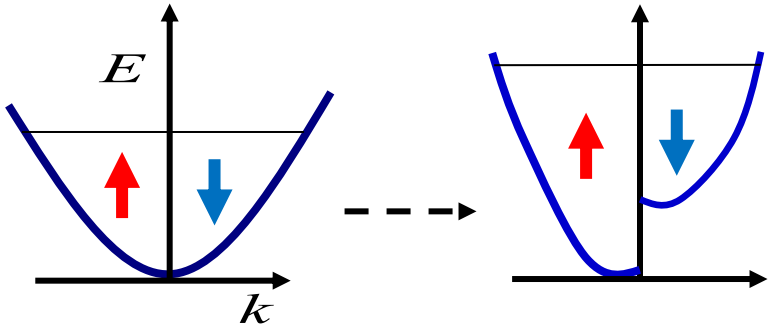


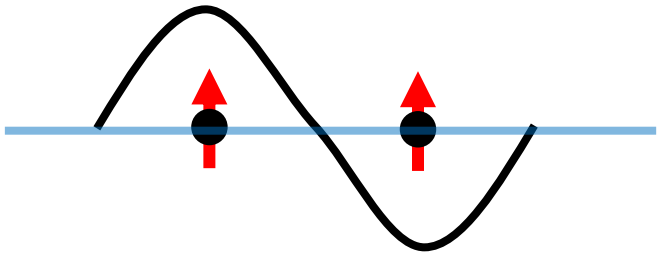
FIG. 11. Various possible types of overlay of e_g orbitals at neighboring centers. a—the overlap of single filled orbitals, which leads to a strong antiferromagnetic exchange interaction; b—the overlap of filled orbitals is zero. A filled orbital and an empty orbital (dashed line) overlap, and the exchange is accordingly ferromagnetic.

Goodenough, J. B. (1955). *Physical Review*, *100*(2), 564.,
Kugel K I, Khomskiĭ D I. Soviet Physics Uspekhi, 1982, 25(4): 231-256.

- Fermi surface -- much harder to form ferromagnetism



FS enlarges, kinetic energy loss



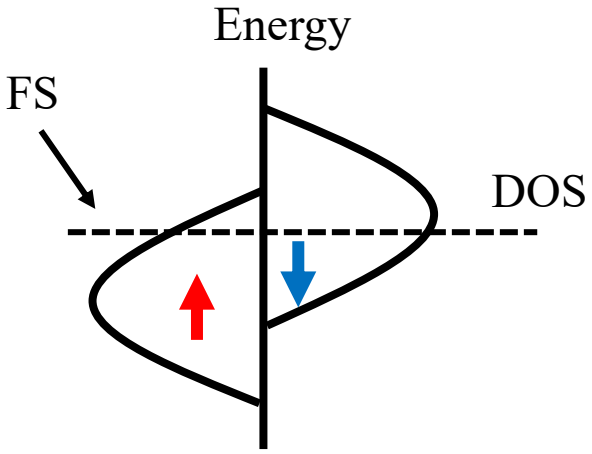
Avoiding overlapping,
interaction energy gains

Theory about ferromagnetism

- Stoner's criterion

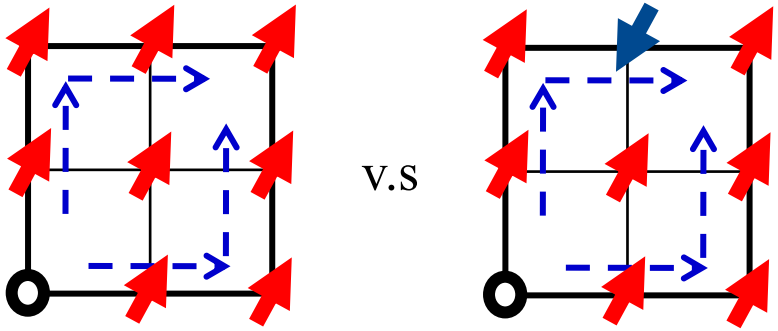
$$U\rho(\varepsilon_F) > 1$$

U -- the on-site interaction;
$\rho(\varepsilon_F)$ -- the density of state on FS



- Nagaoka FM and flat band FM

Configuration space



Momentum space



Nagaoka FM: $N_e = N - 1, U \to \infty$

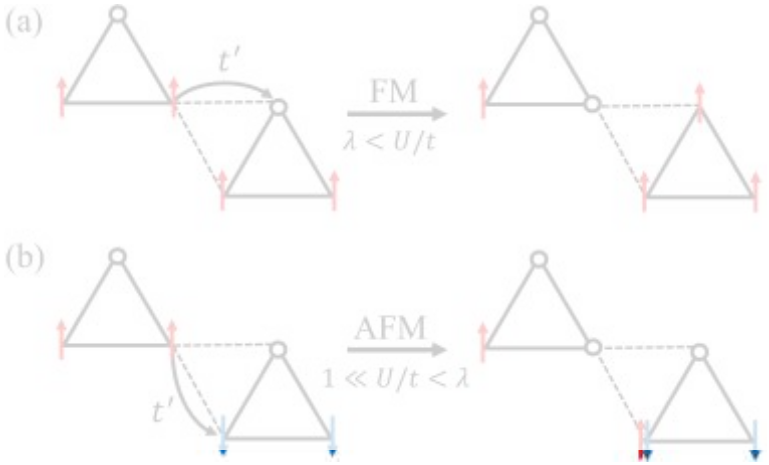
PhysRev.147.392 (1965)

Flat band FM: $W \to 0$

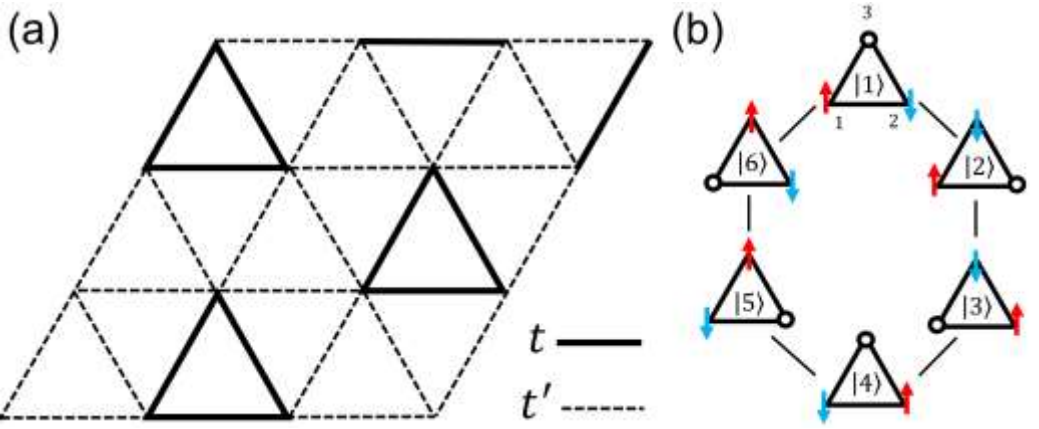
Commun. Math Phys. 158, 341 (1993)

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4. Extension

Nagaoka's ferromagnetism

Hamiltonian:

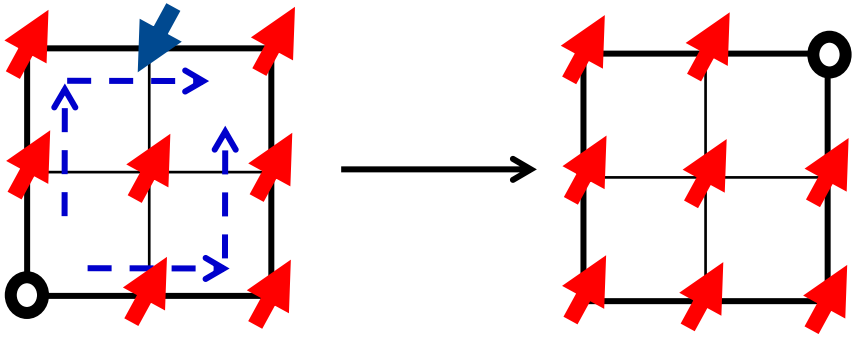
$$H = t\sum_{\langle ij\rangle}\left(c_{i\sigma}^{+}c_{j\sigma} + h.c.\right) + U\sum_{i} n_{i\uparrow}n_{i\downarrow}$$

s.t.

1) Almost half-filling with exactly one hole
2) $U \to \infty, t > 0$
3)Transitivity condition

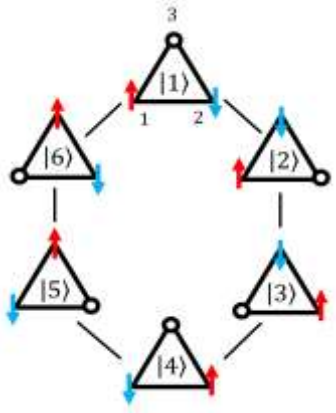
then

The GS is non-degenerate and fully polarized – Ferromagnetism occurs !

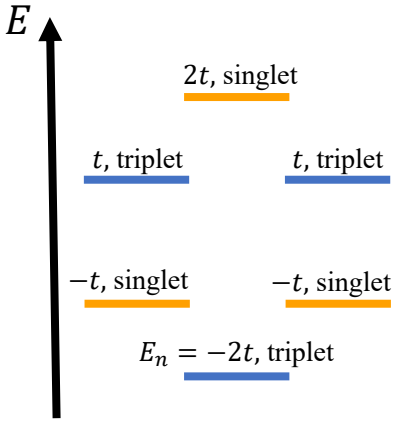


Toy model (trimer)

Configuration space
$S_z = 0$ sector



Momentum space



$$E_n = -2t\cos\left(n\frac{\pi}{3}\right), n = 0, \pm1, \pm2,3$$

$$|GS\rangle \propto \left(|1\rangle + |4\rangle \right)$$

+ other two possible permutations

Trimerized triangular lattice (TTL)

We consider the following Hamiltonian H consisting of two parts:

- Intra-trimer part:

$$H_0 = t\sum_{\langle ij\rangle}\left(c_{i\sigma}^{+}c_{j\sigma} + h.c.\right) + U\sum_{i} n_{i\uparrow}n_{i\downarrow}$$

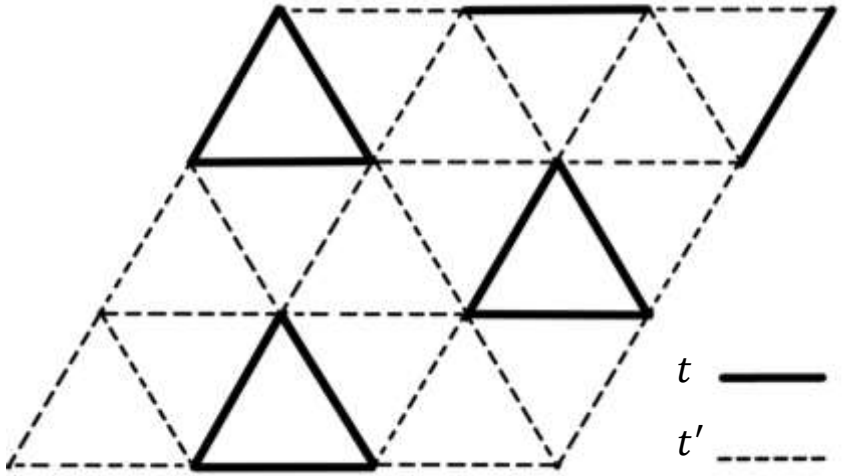
- Inter-trimer part:

$$H' = t'\sum_{\langle\langle i'j'\rangle\rangle}\left(c_{i'\sigma}^{+}c_{j'\sigma} + h.c.\right)$$

$\langle ij\rangle$ is the intra-trimer bonds
$\langle\langle ij\rangle\rangle$ is the inter-timer bonds

FM occurs at

1. 1/3-filling
2. $U \to \infty$
3. $t' \ll t$



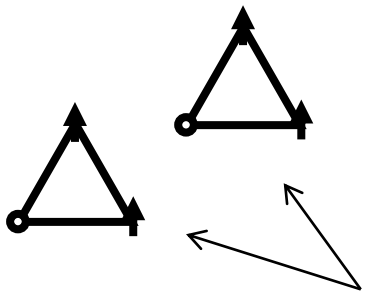
Material: $LiVO_2$

$t > 0$ is important for establishing FM

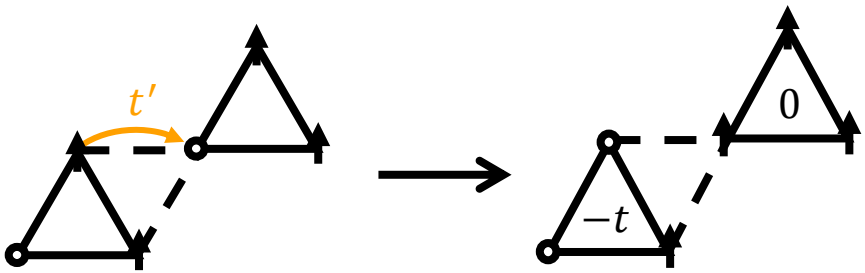
FM coupling between two trimers

FM occurs at

1. 1/3-filling
2. $U \to \infty$
3. $t' \ll t$



Triplet, $S = 1$



According to SU(2) symmetry, the Hilber space is separated into three sectors:

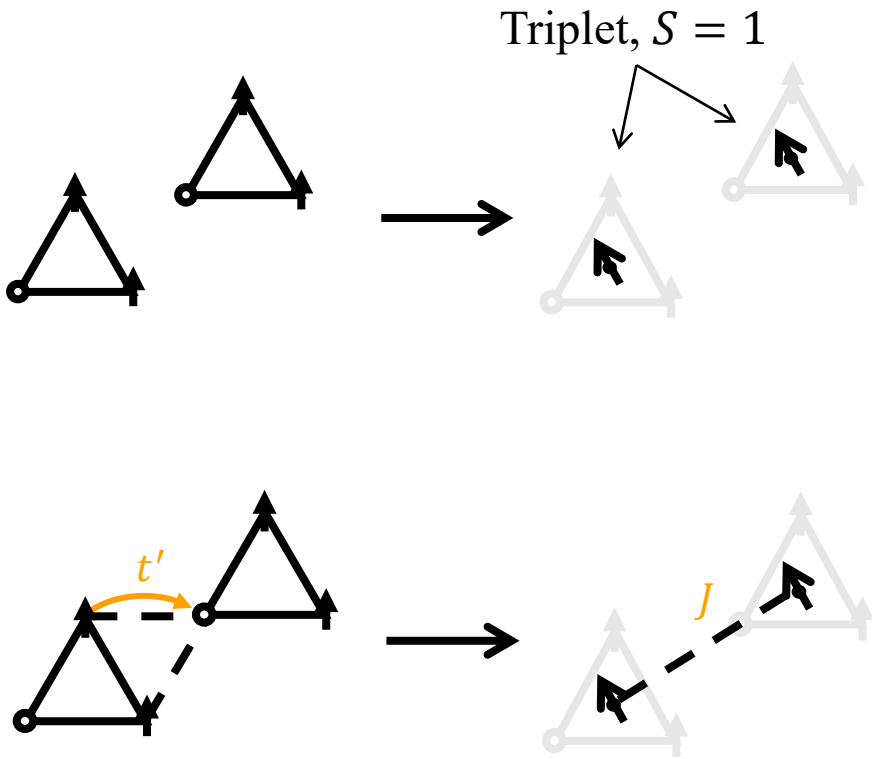
$$1 \otimes 1 = 2 \oplus 1 \oplus 0$$

- The energy gain of original model could be calculated by perturbation theory as $t' \ll t$ is assumed, which are $-\frac{10t'^2}{27t}, -\frac{6t'^2}{27t}, -\frac{4t'^2}{27t}$.

Effective FM Heisenberg model

FM occurs at

1. 1/3-filling
2. $U \to \infty$
3. $t' \ll t$



According to SU(2) symmetry, the Hilber space is separated into three sectors:

$$1 \otimes 1 = 2 \oplus 1 \oplus 0$$

- The energy gain of original model could be calculated by perturbation theory as $t' \ll t$ is assumed, which are $-\frac{10t'^2}{27t}, -\frac{6t'^2}{27t}, -\frac{4t'^2}{27t}$.
- An analogy between the model and Heisenberg model is established by :

$$H = H_0 + H'$$

$$\Rightarrow H_{eff} = J \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j + Const$$

The energy gain of effective Heisenberg model is easy to calculate, which are $-J, J, 2J$

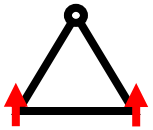
- The effective coupling is determined as

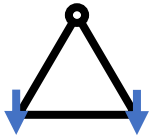
$$J = -\frac{2t'^2}{27t} < 0$$

Physical picture

$$E^{(2)} = \sum_m \frac{|\langle m|H'|\psi\rangle|^2}{E^{(0)} - E_m^{(0)}}$$

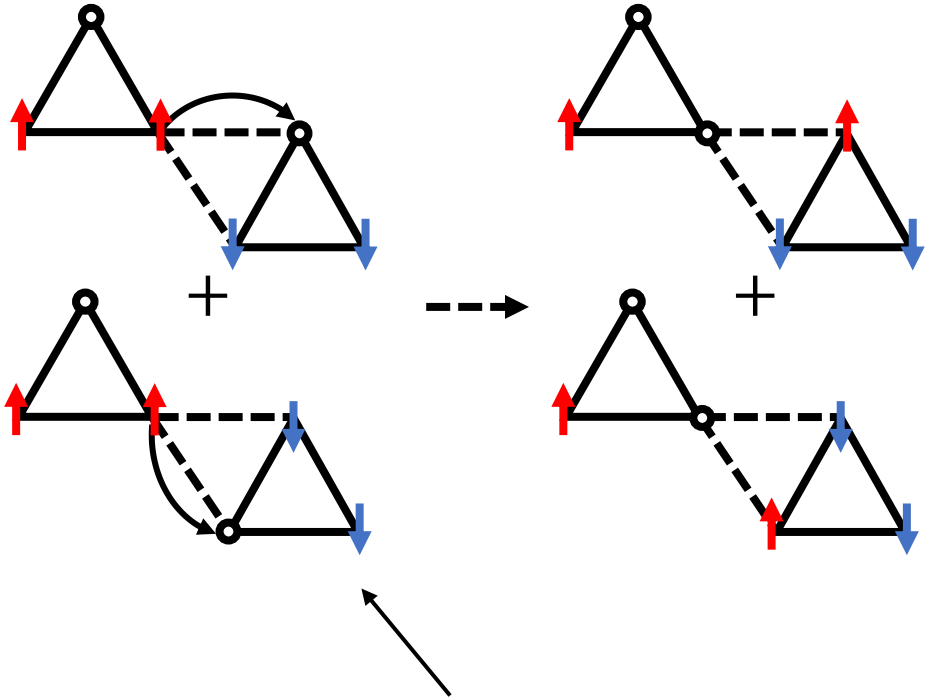
Initial states

$|S = 1, S_z = 1\rangle \propto$  + other two possible permutations

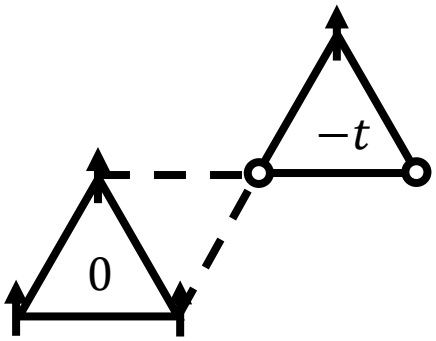
$|S = 1, S_z = -1\rangle \propto$ 

+ other two possible permutations

AFM channel



Energy of intermediate states

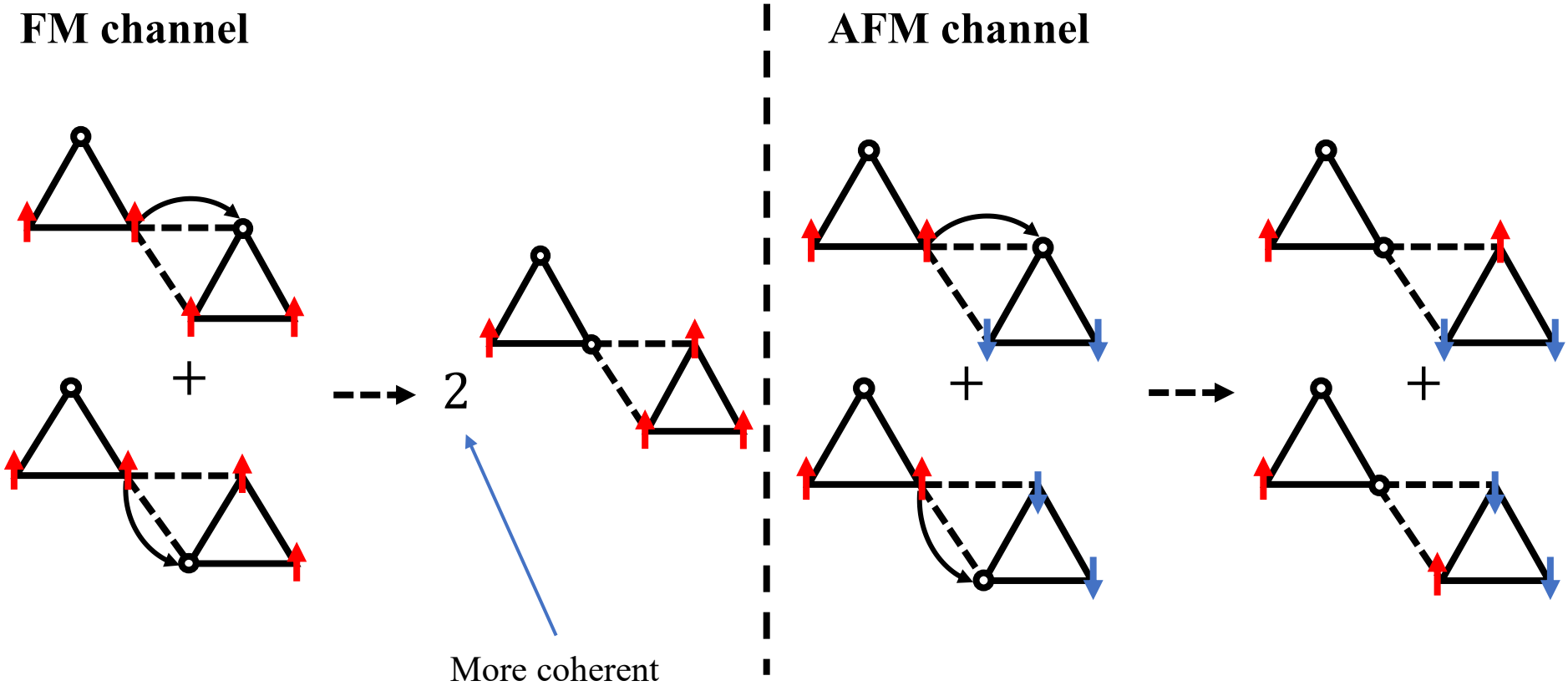


Energies of all intermediate states are $-t$

Two explicit configurations belongs to the superposition of initial state

FM channel v.s. AFM channel

$$E^{(2)} = \sum_m \frac{|\langle m|H'|\psi\rangle|^2}{E^{(0)} - E_m^{(0)}}$$

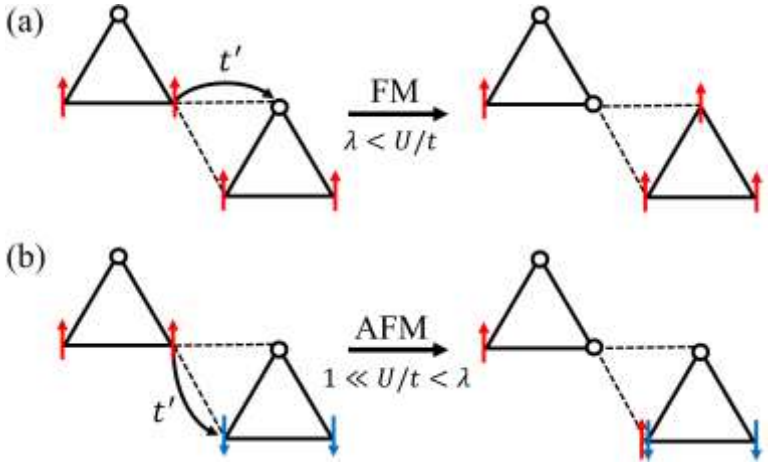


$$H = H_0 + H' \Rightarrow H_{eff} = -\frac{2t'^2}{27t} \sum_{\langle ij \rangle} (\vec{S}_i \cdot \vec{S}_j + 4)$$

1.The origin of ferromagnetism – interplay between kinetic energy and interaction

2.From Nagaoka ferromagnetism to kinetic energy driven ferromagnetic insulator

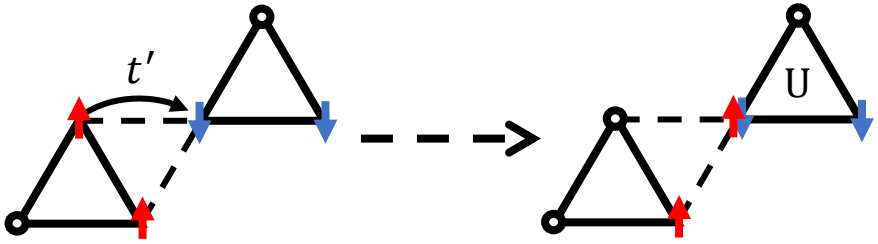
3.Competition between AFM and FM



4. Extension

Interaction-tuning FM-AFM transition

When $t \ll U < \infty$, intermediate states with double occupancy matters



2nd perturbation theory gives:

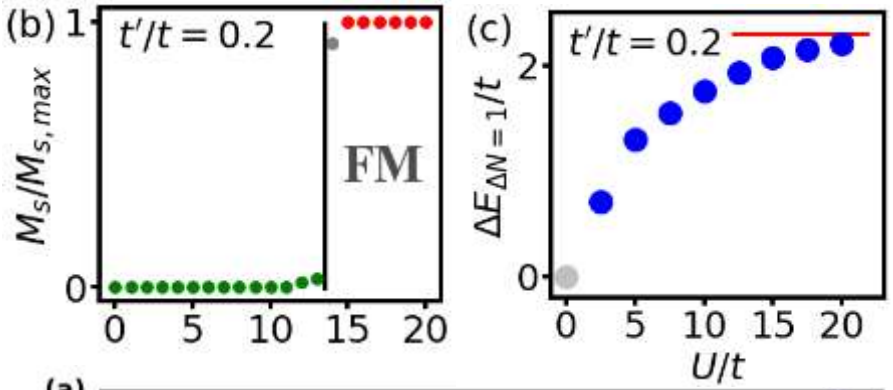
$$J = -\frac{2}{27}\frac{t'^2}{t} + \frac{62}{81}\frac{t'^2}{U} \Rightarrow U_c \approx 10.3$$

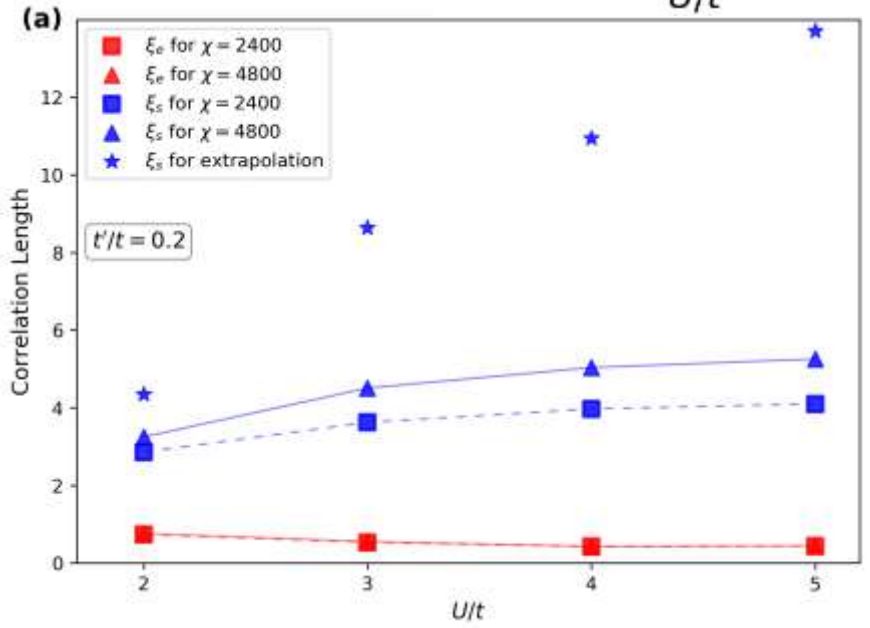
FM channel
Kinetic energy dominant

AFM channel
Interaction dominant

DMRG simulation

Size: $6 \times 4 \times 3$

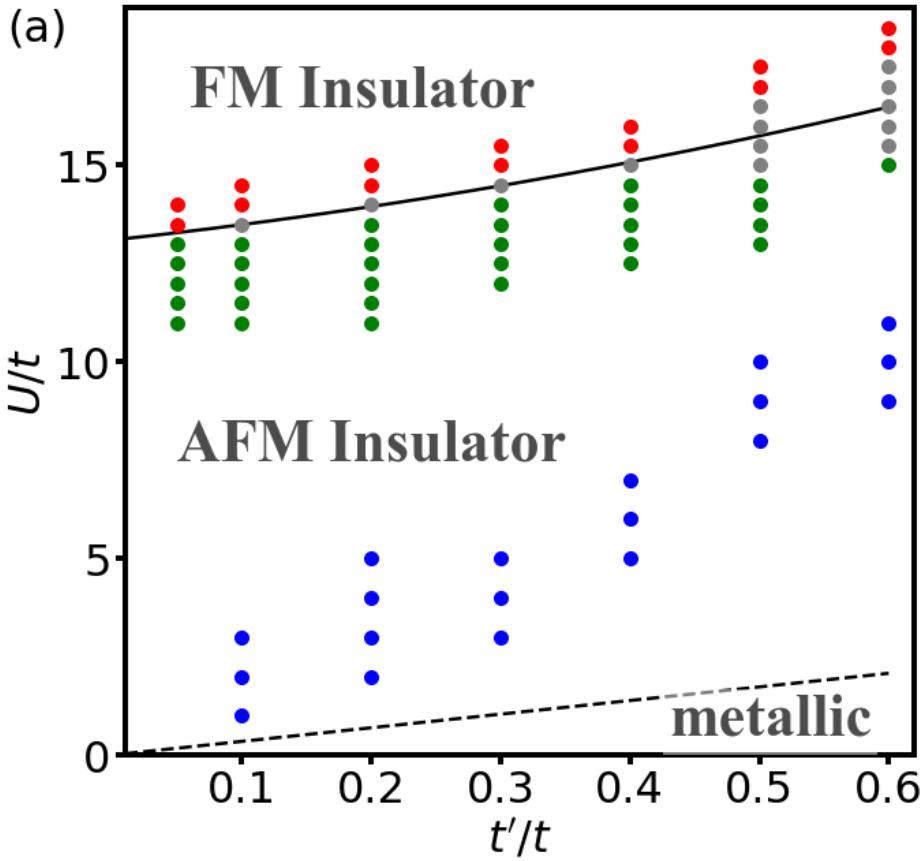


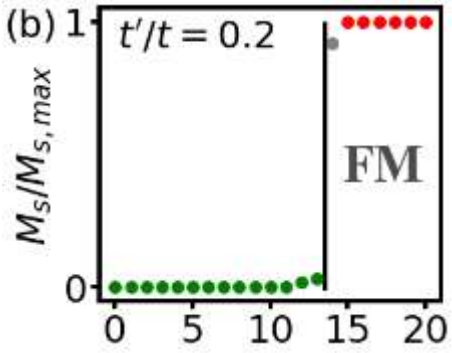


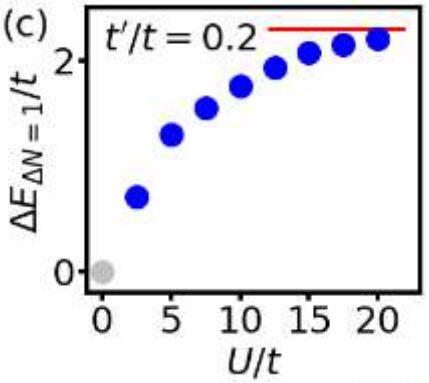
Interaction-tuning FM-AFM transition

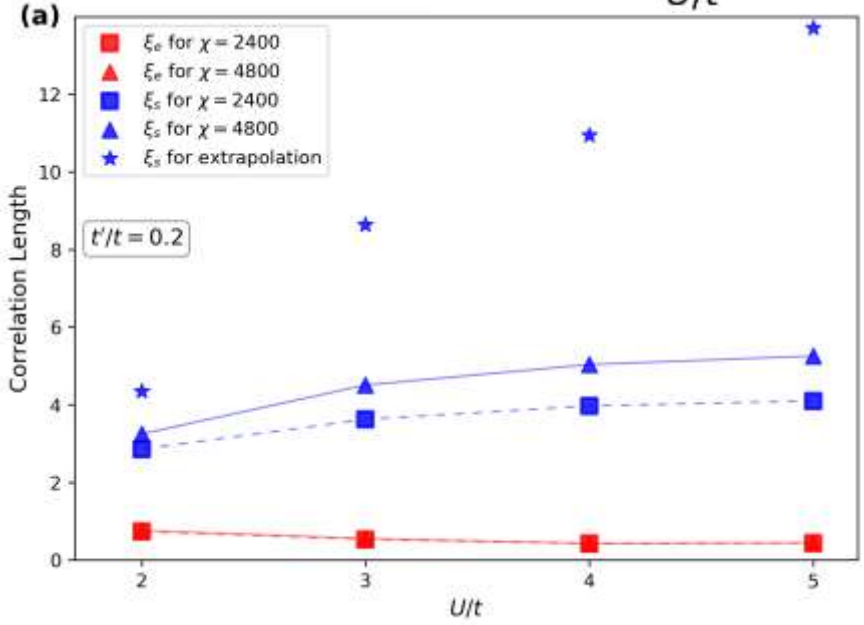
DMRG simulation

Size: $6 \times 4 \times 3$





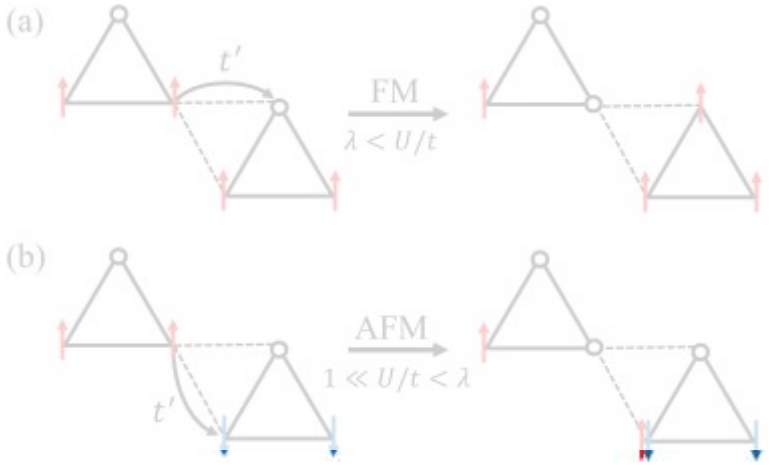




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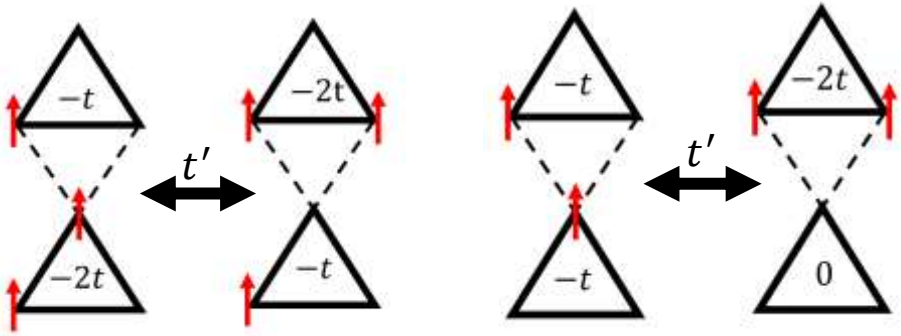
3.Competition between AFM and FM



4. Extension

Adjusting filling factor

When filling factor is reduced lower than 1/3, unperturbed ground states contain the following configuration:



1/4 - filling

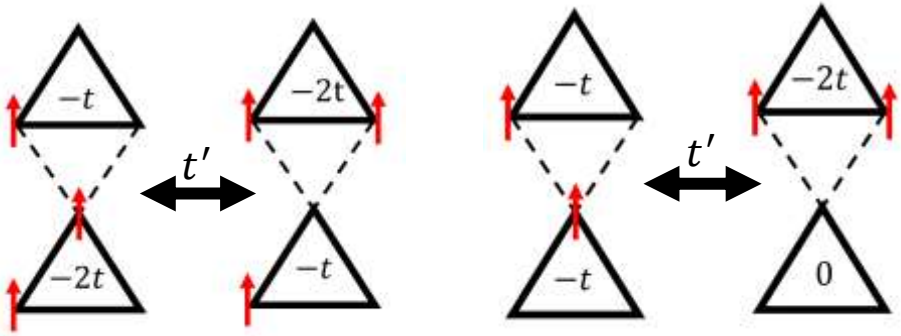
1/6 - filling

There exists gapless excitation states for FM coupling

Due to the coherence of FM channel, the system should turn out to be **ferromagnetic**.

Adjusting filling factor

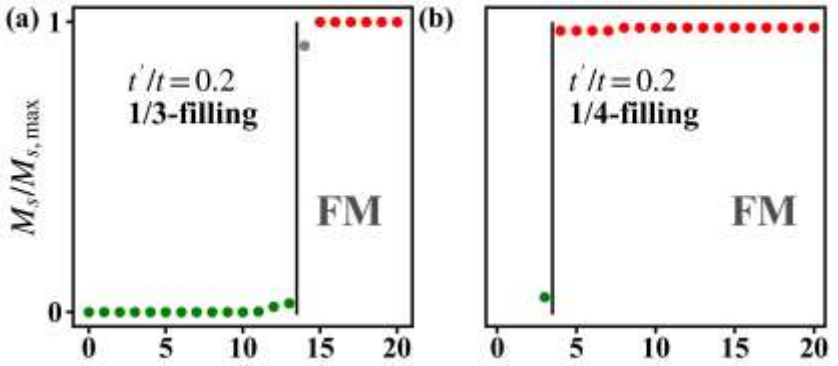
When filling factor is reduced lower than 1/3, unperturbed ground states contain the following configuration:

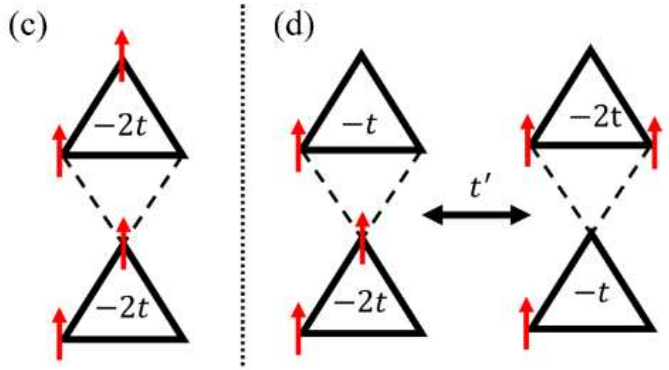


1/4 - filling

1/6 - filling

DMRG simulation





Due to the coherence of FM channel, the system should turn out to be **ferromagnetic**.

Threading flux

Hopping terms carry phase

$$t \to te^{\pm i\frac{\phi}{3}}$$

$$t' \to t'e^{\pm i\frac{\phi}{3}}$$

Eigen-energy of single trimer shifts

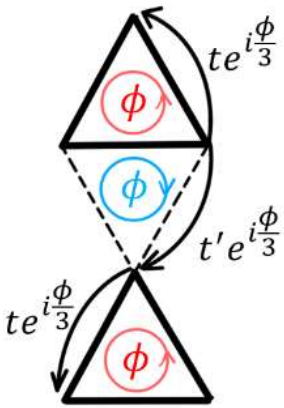
$$E_n = -2t\,cos\left(k_n + \frac{\phi}{3}\right)$$

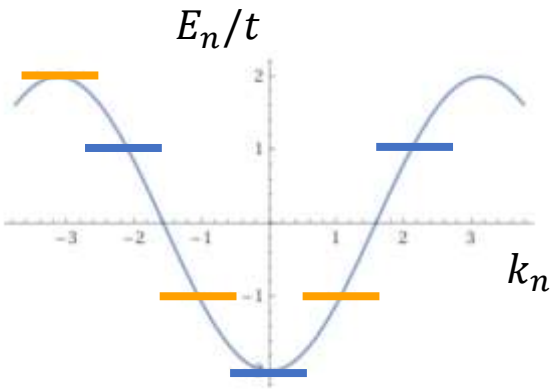
where $k_n = n\frac{\pi}{3}$

FM is robust to small flux ($\phi < \frac{\pi}{2}$)

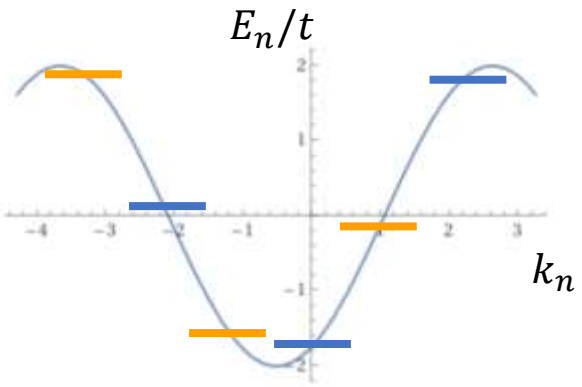
$$J(\phi) = -\frac{t'^2}{18t}\left(\frac{\cos\frac{2\phi}{3}\cos\frac{\phi}{3}}{\cos\left(\frac{\phi}{3}+\frac{\pi}{6}\right)\cos\left(\frac{\phi}{3}-\frac{\pi}{6}\right)}\right)$$

$$\approx J\left(1 - \frac{7}{54}\phi^2\right)$$





$+\phi/3$



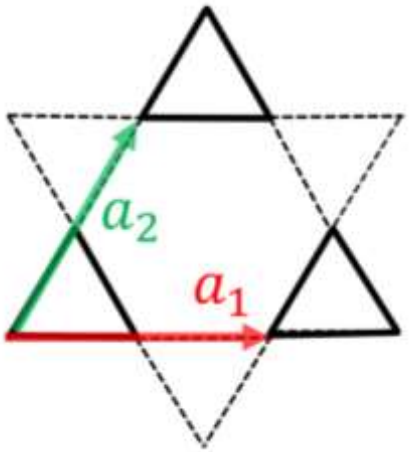
Trimerized Kagome lattice

Effective coupling by 2nd perturbation theory

$$H = H_0 + H'$$

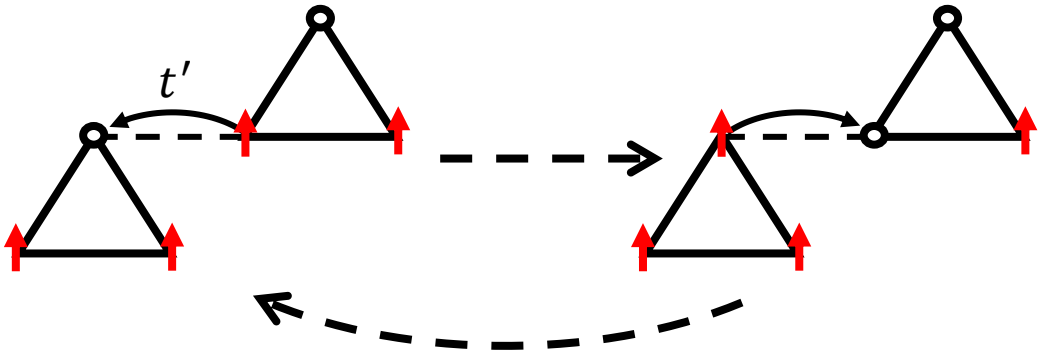
$$\Rightarrow H_{eff} = J \sum_{\langle ij \rangle} (S_i \cdot S_j + 4),$$

where $J = \frac{44}{81} \frac{t'^2}{U}$



Nb_3Cl_8

Physical picture (FM channel)



$$H \propto \vec{S}_1 \cdot \vec{S}_2$$

$$= S_1^z S_2^z + \frac{1}{2}(S_1^+ S_2^- + S_1^- S_2^+)$$

which breaks SU(2) symmetry !

Summary

- We discover a FM insulating state, which is established in the trimerized triangular lattice at the 1/3-filling with $t > 0$ in the regime of $U \gg t \gg |t'|$.
- At $U/t \to \infty$, only the FM superexchange exists. A novel mechanism is proposed for ferromagnetic insulators: this mechanism is primarily driven by kinetic energy reduction.
- As U/t becomes finite, both the FM and AFM superexchanges contribute, and the former wins over the latter around $U/t \gtrsim 15$.
- Our model unifies ferromagnetic (FM) and antiferromagnetic (AFM) phases within a single framework, governed by the competition between kinetic energy-driven FM exchange (dominant at very large U/t) and potential energy-driven AFM exchange.
- Ferromagnetic ground state is robust under lowering filling factor or threading flux
- The trimerized Kagome lattice only has AFM superechange.

Thank you for listening !

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THANKS FOR YOUR ATTENSION!



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